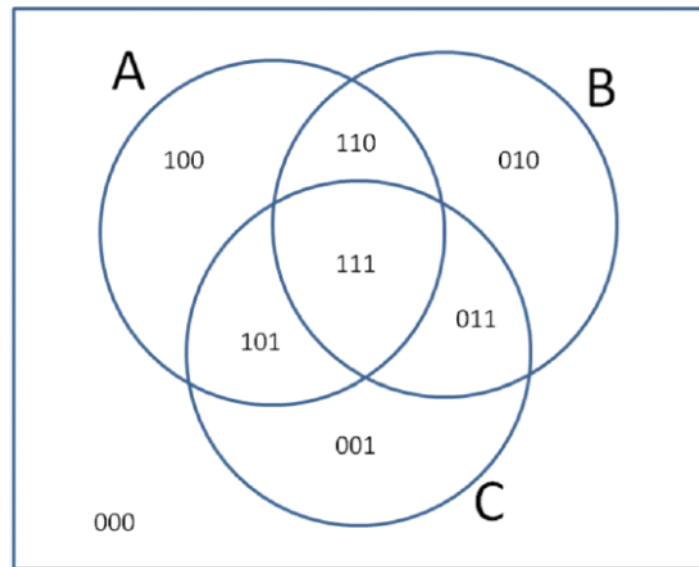

COMP 2121
DISCRETE
MATHEMATICS

Lecture 9



Chapter 3 - Set Theory

Basic Definitions

This chapter introduces the structures of set and ordered set and illustrates them with examples. Together with logic and quantifiers, it provides a strong base for set properties discussion and introduces to the notation of formal languages (computer languages).

Definition: *A set is a collection of objects.*

If a set is finite and not “too large”, we can describe it by listing the elements.

Example 1. $A = \{1, 2, 3, 4\}$ $C = \{1, 2, 3\}$ $|A| = 4$
 $B = \{\}$ $|B| = 0$
 $C = \{1, 2, 3, 1, 2\}$ $|C| = 3$
 $D = \{1, 2, 3, \{1\}, \{1, 2\}\}$ $|D| = 5$

***Note that $\{1\} \neq 1$

For a finite set A , $|A|$ denotes the number of elements of A and is referred to as the *cardinality*, or *size*, of A . For example, set $A = \{1, 2, 3, 4, 7, 10\}$ has a cardinality 6; another words $|A| = 6$.

$\in$ $x \in A$ $x \notin B$ $3 \in A$ $5 \notin A$

The order in which the elements are listed is irrelevant, as is the fact that some elements may be listed more than once.

$x \in A$ expresses that the element x is in set A

Sets Defined by Property

If the set is infinite or too big, the set is described by defining a property. For example:

$A = \{x \in S \mid P(x)\}$ reads as “A is the set of all x in S such that $P(x)$ is true”

Example 2. $A = \{x \in \mathbb{R} \mid -7 < x < 2\}$

$$B = \{x \in \mathbb{Z} \mid -7 < x < 2\}$$

$$C = \{x \in \mathbb{Z}^+ \mid -7 < x < 2\}$$

$$C = \{1\}$$

$\mathbb{R}$ - Real numbers

$\mathbb{Z}$ - set of integers

$\mathbb{Z}^+$ - set of positive integers

Definition: If A and B are sets, A is called a subset of B , written $A \subseteq B$, if and only if, every element of A is also an element of B .

$$A \subseteq B \Leftrightarrow \forall x, \text{ if } x \in A \text{ then } x \in B$$

$$\subseteq ; \leq$$

*** Note distinction between $\in$ and $\subseteq$

It follows that

$$A \not\subseteq B \Leftrightarrow \exists x, x \in A \text{ and } x \notin B$$

$$\text{let } A = \{\}\$$

$$\{\} \not\subseteq B \Leftrightarrow \exists x, x \in \{\} \text{ and } x \notin B$$

* empty set is a subset of every set *

$$A = \{1, 2\}$$

$$B = \{1, 2, 3\}$$

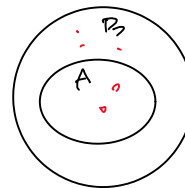
$$C = \{1, 2\}$$

$$A \subseteq B$$

$$A \subseteq B \checkmark$$

$$A \subseteq C \checkmark$$

$$B \subseteq C \times$$



$$A \subset B \checkmark$$

$$A \subset C \times$$

$$B \subset C \times$$

$$A \subseteq B$$

$$A \leq B$$

$$A \subset B$$

$$A < B$$

Definition: Let A and B be sets. A is a proper subset of B if, and only if, every element of A is in B but there is at least one element of B that is not in A .

$$A \subset B \Leftrightarrow \forall x, \text{ if } x \in A \text{ then } x \in B \text{ and } A \neq B$$

Sets can be represented with Venn diagrams:

$$U = \{1, 2, 3, 4, 5\}$$

$$A = \{1, 2, 3\}$$

$$B = \{3, 4\}$$

$$A = \{3\}$$

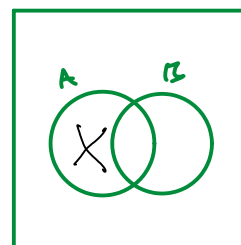
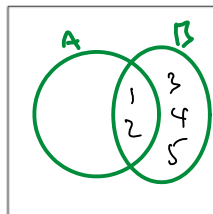
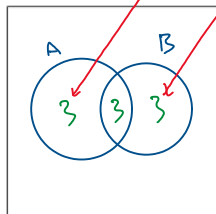
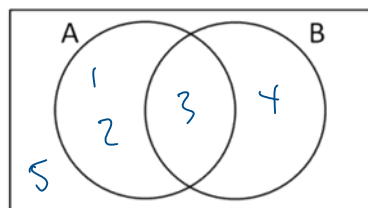
$$B = \{3\}$$

$$A = \{1, 2\}$$

$$B = \{1, 2, 3, 4, 5\}$$

$$A \subseteq B$$

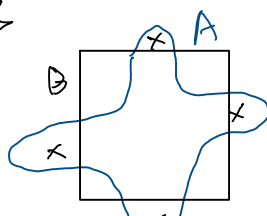
$$A \subseteq B$$



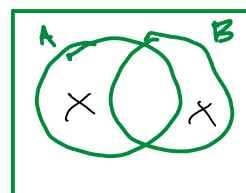
Definition: Given sets A and B , A equals B ($A = B$), if, and only if, every element of A is in B and every element in B is in A .

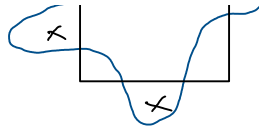
$$A = B \Leftrightarrow A \subseteq B \text{ and } B \subseteq A$$

$$A \subseteq B$$



$$A = B$$





➤ Power Set:

$$\mathcal{P}(A)$$

Definition: Given a set A , the power set of A , denoted by $\mathcal{P}(A)$, is the set of all subsets of A .

For all integers $n \geq 0$, if a set A has n elements then $\mathcal{P}(A)$ has 2^n elements.

Example 3. Find the power set of $\{x, y\}$.

$$A = \{x, y\}$$

$$\mathcal{P}(A) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$$

$$B = \{1, 2, 3\}$$

$$\mathcal{P}(B) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

$$|\mathcal{P}(B)| = 8$$

$$|\mathcal{P}(C)| = 2^{|C|}$$

Set Operations

➤ Operations on Sets

Consider A and B to be subsets of a universal set U

1. The union of A and B ($A \cup B$) is the set of all elements in U such that x is in A OR x is in B.
2. The intersection of A and B ($A \cap B$) is the set of all elements in U such that x is in A AND x is in B.
3. The difference B minus A ($B - A$) is the set of all elements in U such that x is in B AND x is not in A.
4. The symmetric difference $A \Delta B$ is the set of all elements in U such that x is in A OR x is in B, but not in both A and B.
5. The complement of A ($\bar{A}$) is the set of all elements in U such that x is not in A.

Formally:

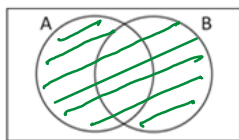
$$A \cup B = \{x \in U \mid x \in A \vee x \in B\}$$

$$A \cap B = \{x \in U \mid x \in A \wedge x \in B\}$$

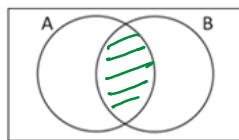
$$A - B = \{x \in U \mid x \in A \wedge x \notin B\}$$

$$A \Delta B = \{x \in U \mid (x \in A \vee x \in B) \wedge x \notin A \cap B\}$$

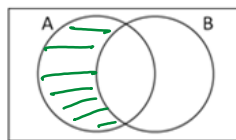
$$\bar{A} = \{x \in U \mid x \notin A\}$$



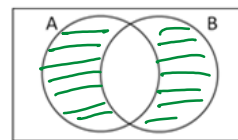
$A \cup B$



$A \cap B$

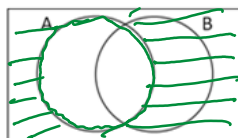


$A - B$

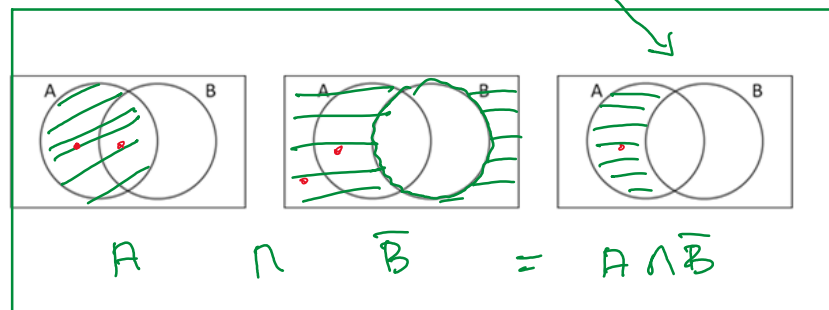


$A \Delta B$

$$A \cap \bar{B} = A - B$$



$\bar{A}$



Example 4

Example 4. With $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ and $C = \{1, 5\}$ we have:

$$A \cup B = \{1, 2, 3, 4\}$$

$$A \cap B = \{2, 3\}$$

$$B \cap C = \{\}$$

$$A \Delta B = \{1, 4\}$$

$$\bar{A} = \{4, 5, 6\}$$

$$A - B = \{1\}$$

$$C - B \subseteq A ?$$

$$\{1, 5\} \subseteq \{1, 2, 3\} \quad \underline{\text{no}}$$

$$B \cap C \subseteq \overline{A \Delta B} \Delta C \quad \underline{\text{yes}}$$

Set Properties

➤ Properties of Sets

A statement regarding a relation between sets can have a value of true or false. Note that the Laws of Set Theory are closely related to the Laws of Logic.

The Laws of Set Theory

For any sets A , B , and C taken from a universe $\mathcal{U}$

- | | |
|---|---------------------------------|
| 1) $\overline{\overline{A}} = A$ | Law of <i>Double Complement</i> |
| 2) $\overline{A \cup B} = \overline{A} \cap \overline{B}$
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ | <i>DeMorgan's Laws</i> |
| 3) $A \cup B = B \cup A$
$A \cap B = B \cap A$ | <i>Commutative Laws</i> |
| 4) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ | <i>Associative Laws</i> |
| 5) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ | <i>Distributive Laws</i> |
| 6) $A \cup A = A$
$A \cap A = A$ | <i>Idempotent Laws</i> |
| 7) $A \cup \emptyset = A$
$A \cap \mathcal{U} = A$ | <i>Identity Laws</i> |
| 8) $A \cup \overline{A} = \mathcal{U}$
$A \cap \overline{A} = \emptyset$ | <i>Inverse Laws</i> |
| 9) $A \cup \mathcal{U} = \mathcal{U}$
$A \cap \emptyset = \emptyset$ | <i>Domination Laws</i> |
| 10) $A \cup (A \cap B) = A$
$A \cap (A \cup B) = A$ | <i>Absorption Laws</i> |

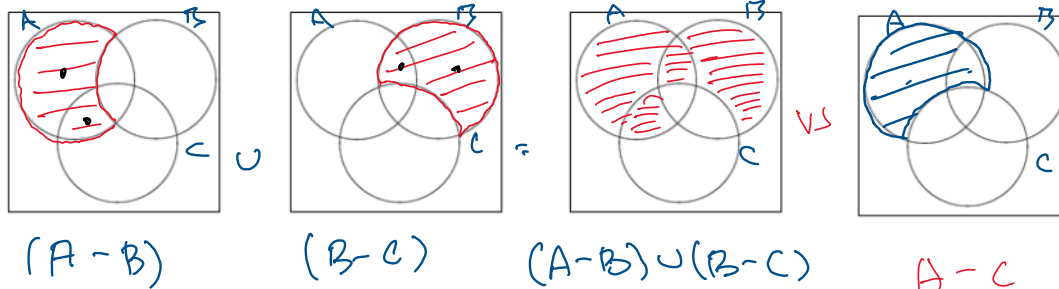
Disproving/finding special examples

Another method for proving set properties is by using Venn diagrams. Note: As the number of sets increases the loss of symmetry in the diagrams is unavoidable.

Example 5. a) Use Venn's diagrams to show that the following identity is false:

$$(A - B) \cup (B - C) = A - C$$

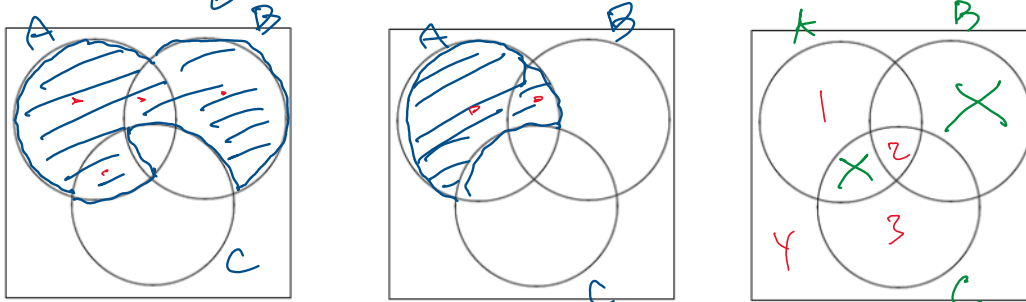
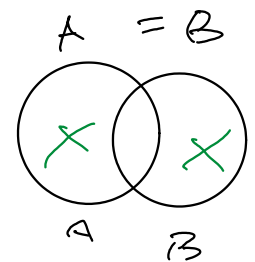
$$\begin{aligned} A &= \{ \} \\ B &= \{ \} \\ C &= \{ \} \end{aligned}$$



b) Even though the given identity is false in general, find one example of nonempty sets A, B and C that satisfies the identity

$$(A - B) \cup (B - C) = A - C$$

Use universe $U = \{1, 2, 3, 4\}$.



$$A = \{1, 2\}$$

$$B = \{2\}$$

$$C = \{2, 3\}$$

$$\begin{aligned} A - B &= \{1\} \\ B - C &= \{ \} \\ (A - B) \cup (B - C) &= \{1\} \\ A - C &= \{1\} \end{aligned}$$

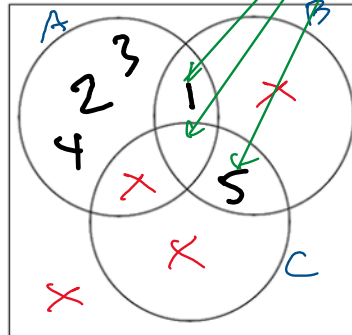
✓

Specific Examples

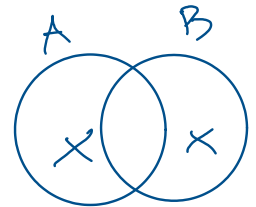
Example 6. Provide one example of nonempty sets A, B and C that satisfies all of the following:

- $(C - (A \cap B)) \subseteq B$
- $(B - C) \subseteq (A \cup C)$
- $(\overline{B \cup C}) \subseteq A$
- $|B| = 2$
- $2 \notin B$

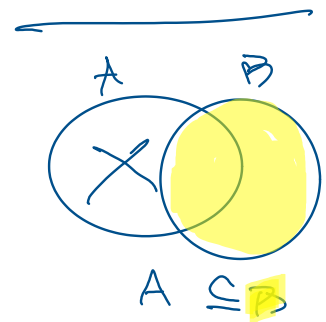
Universe $U = \{1, 2, 3, 4, 5\}$.



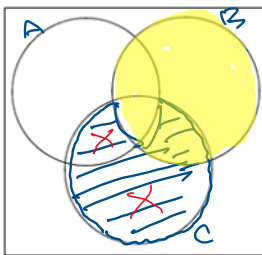
two elements are here
2 is NOT here



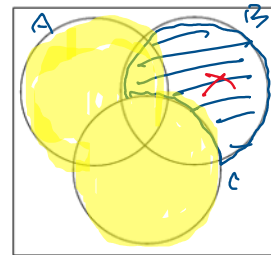
$A = B$



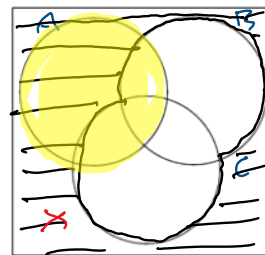
$A \subseteq B$



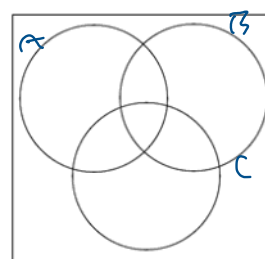
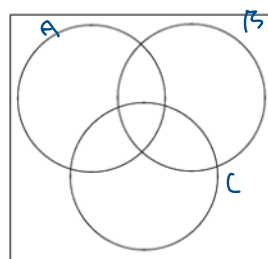
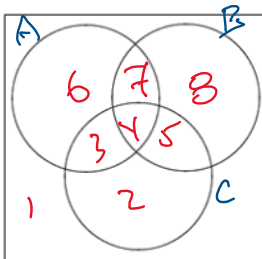
$C - (A \cap B) \subseteq B$



$(B - C) \subseteq (A \cup C)$



$\overline{B \cup C} \subseteq A$

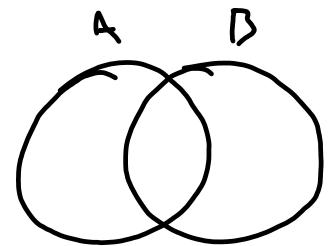


Write your answer:

$A = \{1, 2, 3, 4\}$

$B = \{1, 5\}$

$C = \{5\}$



$A \subseteq B$